

AI Search Framework: Experiment 1 (E1) Empirical Evaluation Report

Fixed 5–8B SLM Pool (No Fine-Tuning) vs. 4-Tier Non-Fine-Tuned Baseline Ladder

Repository: dixitabhi1/SLM_PROJECT (Branch: exp/mentor-protocol-e1)	Benchmark Cohort: 8 Canonical Two-Domain Queries (100% Taxonomy Coverage)
Evaluation Methodology: Symmetrical Double-Blind Judging (64 trials, 16/tier)	Fairness Status: Pre-Flight Verified ($\sum P_{\text{ext}\{SLM\}} = 15.64 \text{ ext}\{B\} < P_{\text{ext}\{Baseline\}}$ across all 4 tiers)

1. Executive Summary & Experimental Objectives

The objective of Experiment 1 (E1) under the Mentor Experiment Protocol is to establish an unperturbed empirical control baseline for the all-SLM architecture prior to targeted parameter adaptation. The system tests whether a fixed, non-fine-tuned pool of Small Language Models in the 5–8B parameter range can approximate or match monolithic Large Language Models across a 4-tier parameter ladder: ~20B (openai/gpt-oss-20b), ~32B (gemini-2.5-flash), ~72B (Qwen/Qwen2.5-72B-Instruct), and ~120B (openai/gpt-oss-120b).

All evaluations were executed under double-blind symmetrical presentation (Forward and Swapped positions) on canonical multi-domain queries spanning all eight technical domain pairings. Symmetrical trials were evaluated independently under both the established 1–5 criteria framework and the 1–10 holistic framework, with equality cases retained as a distinct, first-class Draw outcome.

2. System Architecture & Mandatory Pre-Flight Verification

- SLM Pool Components:** Decomposer (Llama-3.1-8B, 8.03B), Coding Specialist (Qwen2.5-Coder-7B, 7.61B), General/Synthesis Specialist (Llama-3.1-8B, 8.03B), Aggregator (Llama-3.1-8B, 8.03B). All models inference-only.
- Hard Rule 13 (Distinct Models):** All seven participating systems map to genuinely distinct checkpoints on separate endpoints (0 collisions).
- Hard Rule 16(b) (Fairness Pre-Flight Assertion):** For every query, exactly 2 specialists participate ($7.61 \text{ ext}\{B\} + 8.03 \text{ ext}\{B\} = \text{mathbf{15.64 ext}\{B\}}$). In all trials, the baseline parameter count strictly exceeds the combined participating SLM pool: $\sum P_{\text{ext}\{SLM\}} = 15.64 \text{ ext}\{B\} < 20.0 \text{ ext}\{B\} < 32.0 \text{ ext}\{B\} < 72.7 \text{ ext}\{B\} < 120.0 \text{ ext}\{B\}$ (Fairness ratios: 0.78x vs 20B, 0.49x vs 32B, 0.22x vs 72B, 0.13x vs 120B).

3. Overleaf Master Results Table — Experiment 1 (E1)

Baseline Tier	Baseline Model	Params	Framework Mode	SLM Wins	Draws	LLM Wins	Effective SLM Win ($Q_S \geq Q_L$)	Quality Proximity [95% CI]	SLM Score	LLM Score	Mean ΔQ [95% CI]
Tier 1 (~20B)	openai/gpt-oss-20b	20.0B	1–10 Holistic	4 (25.0%)	0 (0.0%)	12 (75.0%)	4 (25.00%)	0.6389 [0.5215, 0.7563]	2.19	4.81	-2.63 [-4.12, -1.13]
			1–5 Criteria	4 (25.0%)	1 (6.2%)	11 (68.8%)	5 (31.25%)	64.06% [52.63%, 75.49%]	2.00	3.10	-1.10 [-1.79, -0.42]
Tier 2 (~32B)	gemini-2.5-flash	32.0B	1–10 Holistic	0 (0.0%)	1 (6.2%)	15 (93.8%)	1 (6.25%)	0.4722 [0.3697, 0.5747]	2.00	6.75	-4.75 [-5.67, -3.83]
			1–5 Criteria	0 (0.0%)	1 (6.2%)	15 (93.8%)	1 (6.25%)	46.35% [37.04%, 55.66%]	1.96	4.10	-2.15 [-2.52, -1.77]
Tier 3 (~72B)	Qwen/Qwen2.5-72B-Instruct	72.7B	1–10 Holistic	0 (0.0%)	1 (6.2%)	15 (93.8%)	1 (6.25%)	0.5833 [0.4601, 0.7066]	2.31	6.06	-3.75 [-4.86, -2.64]
			1–5 Criteria	0 (0.0%)	1 (6.2%)	15 (93.8%)	1 (6.25%)	57.29% [45.39%, 69.19%]	2.04	3.75	-1.71 [-2.18, -1.23]
Tier 4 (~120B)	openai/gpt-oss-120b	120.0B	1–10 Holistic	0 (0.0%)	1 (6.2%)	15 (93.8%)	1 (6.25%)	0.4167 [0.3033, 0.5300]	1.88	7.13	-5.25 [-6.27, -4.23]
			1–5 Criteria	0 (0.0%)	1 (6.2%)	15 (93.8%)	1 (6.25%)	42.71% [32.09%, 53.32%]	1.90	4.19	-2.29 [-2.72, -1.87]

Section 3.1 — Treatment of Draws in Favor of SLMs ($Q_S \geq Q_L$):

Per Section 5 of the Mentor Experiment Protocol, reporting draws separately allows examining the impact of crediting draws in favor of the resource-constrained pipeline. If an entirely local ≤8B pipeline achieves parity with a 20B–120B cloud model, parity represents an architectural victory for the SLM system. Under this decision rule, the non-fine-tuned SLM pipeline achieves an effective win rate of **31.25%** against the ~20B baseline (Criteria framework), and **6.25%** against 32B, 72B, and 120B baselines.

4. Mathematical Formulations & Statistical Methods

Holistic Quality Proximity (1–10 Scale, Mentor Protocol Section 4.2):

$QP_i = 1 - (|Q_{S,i} - Q_{L,i}| / 9.0) \Rightarrow QP_{overall} = (1 / N) \sum [1 - (|Q_{S,i} - Q_{L,i}| / 9.0)]$

Where $Q_{S,i}, Q_{L,i} \in [1, 10]$. Maximum possible difference is $10 - 1 = 9.0$.

95% Confidence Interval:

$CI_{95} = [\bar{x} - t_{0.975, N-1} \times (s / \sqrt{N}), \bar{x} + t_{0.975, N-1} \times (s / \sqrt{N})]$

Where s is sample standard deviation and $t_{0.975, 15} = 2.131$ for N=16 paired trials per baseline tier.

5. Autonomous Audit Loop Diagnostics & Verification

- **Score/Label Concordance: 100.0%** across all 64 trials. Unblinded winner strictly matched higher criteria/holistic score sum.
- **Positional Swap Consistency: 75.0%** agreement vs 20B; **87.5%** agreement vs 32B, 72B, and 120B under forward vs swapped presentation.
- **Zero Synthetic Scores:** 100% of reported values were calculated directly from empirical logs on disk.
- **Data Preservation:** All 10 mandated data fields preserved in `results/mentor_protocol/e1/e1_preserved_data.jsonl`.